

Entry

Periapical Lesions: Diagnosis, Pathophysiology, and Management

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Definition

The term “periapical lesion” refers to a pathological change in the tissues surrounding the apex of a tooth root, defined by its anatomical location rather than a distinct disease entity. Periapical lesions may be of endodontic origin, most commonly resulting from microbial infection of the root canal system following pulp necrosis due to caries, trauma, or other insults, or of non-endodontic origin, such as developmental cysts, benign and malignant odontogenic and non-odontogenic tumors, and fibro-osseous lesions. Accurate diagnosis requires a systematic approach combining patient history, clinical examination, pulp vitality testing, and radiographic assessment; histopathological evaluation is indicated when clinical and radiographic findings are inconsistent or suspicious. The pathophysiology of these lesions involves dynamic interactions between root canal microorganisms and the host immune-inflammatory response. The primary management for endodontic periapical lesions is root canal treatment, which aims to reduce or eliminate root canal microorganisms through mechanical debridement and chemical disinfection. Persistent or extensive endodontic lesions and non-endodontic lesions may require surgical intervention. Molecular and inflammatory biomarkers have been investigated as adjunctive tools for assessing disease activity and prognosis; however, these remain largely investigational and are not yet part of routine clinical practice. Future developments in artificial intelligence, advanced imaging, molecular diagnostics, and personalized therapies may enhance the diagnosis and management of periapical lesions, although further clinical validation is required.

Keywords: apical periodontitis; apicoectomy; biomarker; diagnostic imaging; periapical disease; periapical granuloma



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1. Introduction

Periapical lesions are pathological changes in the tissues surrounding the root apex, most commonly resulting from microbial infection of the root canal system following pulp necrosis secondary to caries, trauma, or other insults [1]. The inflammatory reaction in the periapical tissues is induced and maintained by microorganisms colonizing the necrotic root canal, not by the necrotic pulp tissue itself [2]. Additionally, benign and malignant lesions and developmental anomalies may cause periapical bone defects [3].

Periapical lesions are classified radiographically as radiolucent, radiopaque, or mixed, and further subdivided by odontogenic versus non-odontogenic origin [3].

Periapical lesions are common globally, with a notably higher prevalence in endodontically treated teeth than in untreated teeth [4]. Most lesions are asymptomatic and detected incidentally on radiographs, although some present with pain, swelling, or sinus tracts and can mimic other pathological conditions [5]. Most periapical lesions are inflammatory and arise from endodontic causes, such as granulomas, cysts, abscesses, or periapical scars. A smaller proportion are non-endodontic in origin, including benign tumors, cysts, and malignancies, which necessitate histopathological examination for accurate diagnosis and proper management [6,7].

Periapical lesions are the leading cause of tooth loss, often require endodontic or surgical treatment, and can significantly impact treatment planning, particularly in patients undergoing prosthetic rehabilitation or with systemic health concerns [6]. Misdiagnosis of non-endodontic lesions can lead to inappropriate therapy, highlighting the critical role of biopsy in atypical or suspicious cases [7].

Although numerous studies have investigated periapical lesions, the literature is extensive, scattered across different journals, and sometimes inconsistent. Therefore, a clear and reliable summary is needed. This Entry Paper summarizes current evidence regarding the diagnosis, pathophysiology, imaging, management, biomarkers, and future directions of periapical lesions. Relevant literature was identified through targeted searches of PubMed/MEDLINE, Scopus, and Google Scholar using terms including “periapical lesions,” “apical periodontitis,” “diagnostic imaging,” “cone-beam computed tomography,” “endodontic surgery,” “biomarkers,” and “artificial intelligence.” Priority was given to recent systematic reviews, meta-analyses, clinical studies, and other publications considered relevant to the clinical and biological aspects of periapical disease.

2. Diagnostic Criteria and Imaging Approaches

2.1. Clinical Diagnostic Criteria

The clinical diagnosis of periapical lesions relies on subjective symptoms and objective examination findings [8]. The majority of periapical lesions are asymptomatic and are often identified incidentally during routine dental or endodontic examinations [4]. When present, clinical signs may include localized pain, swelling, sensitivity to percussion or palpation, sinus tract formation, or referred pain to the face or ear [5,9].

Clinical examination (Table 1) plays a central role in evaluating the affected tooth. Percussion and palpation tests help detect localized inflammation [8]. Hypersensitivity to percussion is a strong indicator of acute pain and is more commonly seen in teeth with necrotic pulps, whereas tenderness on palpation is frequently observed in previously treated teeth [9]. Together, these tests help distinguish pulpal from periapical conditions [9,10]. Evaluating tooth mobility and the surrounding periodontal tissues is crucial for assessing lesion severity and the extent of tissue involvement. Increased mobility can signal advanced periapical or periodontal disease and examining periodontal pockets and patterns of bone loss offers further insight into the lesion’s effect on the supporting structures [11].

Pulp sensibility tests, such as thermal (cold and heat) and electric pulp testing, are essential for differentiating vital from non-vital teeth. These tests evaluate the response of nerve fibers rather than the actual pulpal blood flow; thus, a positive response indicates functional nerves but does not necessarily reflect the health of the pulp’s vascular system [12]. In contrast, laser Doppler flowmetry and pulse oximetry directly measure the pulpal blood flow and oxygenation, respectively [13,14]. Therefore, they are the most accurate and clinically useful methods for assessing pulp vitality in patients with periapical lesions. The test cavity method is invasive and irreversible, and its clinical utility is limited

because it does not provide additional diagnostic information beyond what is obtained from thermal and electric pulp sensibility tests [15].

Table 1. Clinical tests for the diagnosis of periapical lesions.

Test Type	Method	Key Notes/Advantages	Limitations	Typical Interpretation
Periapical assessment tests				
Percussion test	Tapping the tooth vertically or horizontally	Detects inflammation of periapical tissues. Is simple, quick and non-invasive	Subjective, may be influenced by patient pain threshold	Positive response commonly indicates acute apical inflammation
Palpation test	Firm pressure applied to periapical area	Detects tenderness or swelling in the region. Simple, non-invasive	Subjective	Tenderness may indicate spread of inflammation into surrounding tissues
Sinus tract evaluation	Tracing sinus tract with gutta-percha or probe	Determines lesion origin and drainage	Invasive, requires careful manipulation	Help localize the causative tooth
Pulp assessment tests				
Pulp sensibility tests	Thermal (cold or heat) or electric pulp testing	Assesses sensory nerve response and helps distinguish vital from non-vital pulp. Quick and widely available	Measures nerve response, not true blood flow	Absence of response suggests pulp necrosis
Pulp vitality assessment (laser Doppler/pulse-oximetry)	Measurement of pulpal blood flow or oxygen saturation	Useful in traumatized or immature teeth	Limited availability and higher cost	Provides direct information about the pulp status
Periodontal assessment tests				
Mobility test	Manual assessment of tooth movement	Evaluates structural involvement of periapical or periodontal tissues	May not detect early lesions. Influenced by periodontal status	Increased mobility may indicate advanced tissue destruction
Periodontal assessment	Probing depth and attachment evaluation	Helps identify combined lesions	Cannot confirm periapical pathology alone	Isolated deep probing may indicate root fracture or combined lesion

A systematic diagnostic workflow is recommended: (1) thorough patient history and clinical examination, including percussion, palpation, and mobility assessment; (2) pulp sensibility and, when available, vitality testing; (3) radiographic assessment (see Section 2.2); and (4) correlation of all findings to guide further imaging or biopsy when indicated. This integrated approach minimizes the risk of misdiagnosis, particularly for non-endodontic lesions presenting as periapical pathology.

2.2. Conventional Radiographic Assessment

Although clinical examination provides crucial initial information, it is often complemented by imaging techniques to fully characterize the lesion. Conventional radiographic techniques (two-dimensional imaging), including periapical and panoramic radiography, are the first-line imaging modalities for evaluating periapical lesions [16]. Periapical radio-

graphs are the standard intraoral imaging modality and provide high spatial resolution for the detection of periapical changes. In contrast, panoramic radiographs offer a broader overview of the jaws but with lower resolution and sensitivity for small lesions [17]. Both modalities are used to classify periapical lesions based on their radiographic appearance as radiolucent, radiopaque, or mixed [3].

Radiolucent periapical lesions appear as dark areas at the apex and are generally associated with bone destruction (Figure 1). Odontogenic radiolucent lesions most commonly include periapical granuloma, radicular cyst, and periapical abscess. However, reliable differentiation between periapical granulomas and radicular cysts cannot be achieved by radiographic examination alone; histopathological examination is required for definitive diagnosis [6,7]. Additionally, some non-odontogenic lesions, such as nasopalatine duct cyst, central giant cell granuloma, and malignancies, are radiolucent [3,18].



Figure 1. Periapical radiographs demonstrating radiolucent periapical lesions. **(Left):** small periapical radiolucency at the root apices suggestive of a periapical granuloma. **(Right):** larger well-defined radiolucency with corticated borders suggestive of a radicular cyst. Radiographic characteristics such as lesion size, border definition, and relationship to the involved tooth may aid differential diagnosis; however, definitive distinction between granuloma and cyst requires histopathological examination because radiographic appearance alone is insufficient.

Radiopaque periapical lesions (Figure 2) present as dense, white areas and are typically bone-forming [19]. Odontogenic radiopaque lesions include condensing osteitis, hypercementosis, ossifying fibroma, and odontoma, whereas non-odontogenic radiopaque lesions include idiopathic sclerosing osteitis and osteoma. The term “osteitis” warrants clarification: condensing osteitis (also termed focal sclerosing osteomyelitis) refers to a localized area of increased bone density at the root apex, representing a reactive bone response to low-grade inflammatory stimulation from a tooth with pulpitis or a necrotic pulp [20]. This should be distinguished from idiopathic osteosclerosis, a similar-appearing radiopacity unrelated to a diseased tooth and considered a developmental variation [20].

Mixed periapical lesions show both radiolucent and radiopaque components. Odontogenic mixed lesions include cemento-osseous dysplasia, calcifying odontogenic cysts, and residual radicular cysts with calcifications, whereas non-odontogenic mixed lesions include fibrous dysplasia and Paget’s disease [3,21].



Figure 2. Periapical radiograph demonstrating a radiopaque periapical lesion suggestive of condensing osteitis. The lesion appears as a localized area of increased bone density adjacent to the root apex and is typically associated with chronic low-grade inflammatory stimulation from a tooth with pulpal inflammation or necrosis.

Conventional radiography is important for initial evaluation and classification but has well-documented limitations: periapical radiographs require approximately 30–50% mineral bone loss before a lesion becomes detectable, and both sensitivity and specificity for detecting periapical pathology are lower than those of three-dimensional imaging [22]. Anatomical superimposition, geometric distortion, and image noise further reduce diagnostic accuracy. Therefore, correlation with clinical findings is essential, and advanced imaging may be warranted when conventional radiography is inconclusive. Histopathological examination remains the gold standard for definitive diagnosis, as it allows direct evaluation of the lesion's cellular and tissue characteristics.

2.3. Advanced Imaging Modalities

The two-dimensional nature of conventional radiography introduces well-documented limitations for periapical assessment as mentioned above. To overcome these limitations, cone-beam computed tomography (CBCT) is increasingly incorporated into endodontic practice [23,24].

CBCT provides three-dimensional volumetric assessment with superior detection of periapical lesions compared with periapical radiographs, particularly for early lesions, missed canals, root fractures, and resorptive defects. It enables accurate evaluation of lesion extent, volume, and proximity to vital anatomical structures such as the inferior alveolar nerve, maxillary sinus, and mental foramen, which is essential for surgical planning [24,25]. The American Dental Association and American Academy of Oral and Maxillofacial Radiology recommend CBCT when conventional radiography is inconclusive or when detailed anatomical information is required, particularly in cases of persistent or recurrent complex endodontic pathoses and for surgical planning [25].

However, CBCT has notable limitations. Effective radiation doses range from approximately 9 to 175 μSv , depending on the field of view and protocol settings, compared with approximately 1.4 μSv for a single periapical radiograph [26]. The ALARA (As Low As Reasonably Achievable) principle should guide all CBCT prescriptions: the smallest field of view (FOV) necessary should be selected (e.g., a limited FOV of 4×4 cm or 5×5 cm for endodontic applications), and exposure parameters should be optimized to minimize dose while maintaining diagnostic image quality [26,27]. Furthermore, CBCT is more costly than

conventional radiography, is not universally available, and image quality may be degraded by metallic artifacts from existing restorations or endodontic materials [27].

2.4. Differential Diagnosis of Periapical Lesions

Accurate differential diagnosis of periapical lesions is fundamental to proper clinical decision-making. Epidemiological data indicate that 94–97% of periapical radiolucencies are inflammatory in nature and of endodontic origin. In addition to periapical abscesses and periapical scars, periapical granulomas (approximately 60%) and radicular cysts (37–39%) account for the majority of cases [6,7].

However, not all radiolucent lesions have an endodontic origin. Non-endodontic radiolucent lesions may mimic common periapical lesions but are frequently associated with vital teeth or present with atypical features. The most common non-endodontic periapical lesions include odontogenic keratocyst (27–37% of non-endodontic lesions), ameloblastoma (up to 25%), dentigerous cyst, and central giant-cell granuloma [6,7,28]. Importantly, a previous study showed that up to 27% of teeth associated with odontogenic cysts are vital, indicating that root canal therapy would be unwarranted in these cases [29]. Generally, features such as lesion diameter ≥ 2 cm, clinical swelling, non-rounded or irregular radiographic margins, unusual location, cortical expansion, root displacement, or lack of correlation with the pulpal status suggest a non-endodontic etiology [6,18].

Radiopaque periapical lesions represent a distinct diagnostic category that requires differentiation from the more common radiolucent inflammatory lesions of endodontic origin. These lesions may be reactive, developmental, or neoplastic in nature, and accurate diagnosis depends on careful integration of clinical findings, pulp vitality test results, and imaging characteristics [30]. Conditions such as condensing osteitis, cementoblastoma, ossifying fibroma, cemento-osseous dysplasia, calcifying odontogenic cyst, and fibrous dysplasia may involve the periapical region and require differentiation from inflammatory disease [18].

Malignant neoplasms constitute 3.7–5.8% of all non-endodontic periapical lesions, with intraosseous squamous cell carcinoma and metastatic lesions being the most common; these may present with swelling and pain, mimicking endodontic pathosis [28].

Traumatic dental injuries are an important cause of periapical lesions, distinct from the more common pathway of carious pulp exposure. Following trauma, the dental pulp may suffer ischemic injury due to disruption of the neurovascular supply at the apical foramen [31]. In mature teeth, revascularization is unlikely, and the necrotic pulp may remain uninfected for an extended period; however, bacteria eventually reach the pulp space through dentinal cracks, fracture lines, or defective restorations, leading to infection of the root canal system and subsequent periapical inflammation. In immature teeth with open apices, revascularization is more favorable and may allow pulp recovery if microbial contamination does not occur [31].

Systemic diseases may also manifest as periapical radiolucencies, necessitating careful differentiation from inflammatory endodontic lesions. Hematologic malignancies, particularly lymphoma and leukemia, are among the most clinically significant mimickers, frequently presenting as periapical radiolucencies associated with vital teeth and often misdiagnosed as endodontic pathology, leading to delayed diagnosis and inappropriate treatment [32]. Hyperparathyroidism can lead to brown tumors, osteolytic lesions caused by excessive parathyroid hormone-mediated bone resorption, which may radiographically resemble odontogenic cysts or periapical granulomas [33].

2.5. Indications for Biopsy

Biopsy and histopathological examination are indicated when clinical and radiographic findings are inconsistent with a straightforward endodontic diagnosis (Figure 3, Table 2). Specific indications include: lesions that do not respond to adequate endodontic treatment or retreatment; lesions associated with vital teeth and no identifiable endodontic cause; atypical radiographic features such as irregular borders, cortical destruction, or root resorption not explained by inflammatory disease; rapid growth or progressive enlargement; clinical suspicion of neoplastic or systemic disease; and any lesion removed surgically, to confirm the diagnosis and exclude malignancy [6].

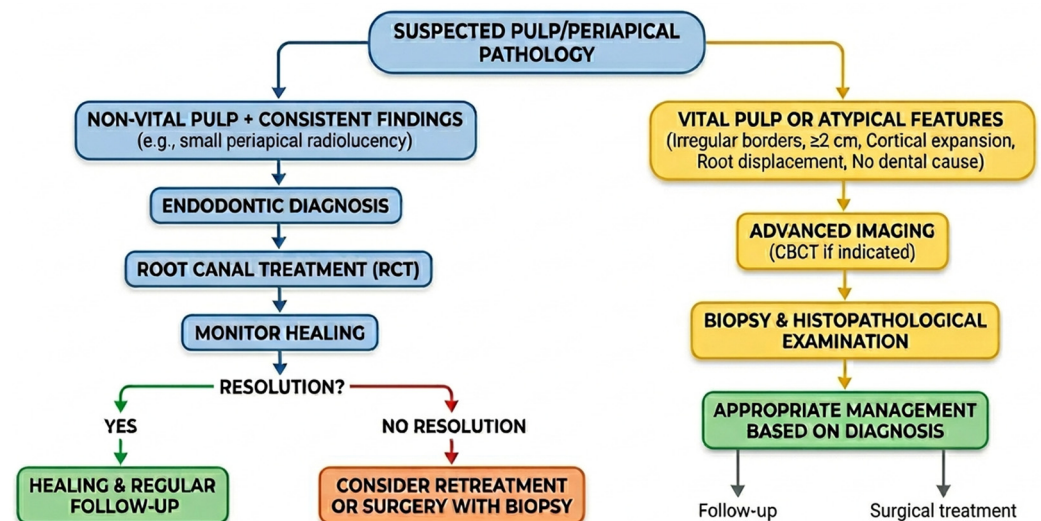


Figure 3. Suggested diagnostic and treatment workflow for suspected pulp and periapical pathology. Teeth with non-vital pulps and typical radiographic findings are generally managed with conventional root canal treatment and follow-up. Lesions with atypical clinical or radiographic features, lesions associated with vital teeth, or lesions unresponsive to treatment require advanced imaging, biopsy, histopathological evaluation, and diagnosis-specific management.

Table 2. Common examples of differential diagnostic features of periapical lesions and findings warranting histopathologic evaluation.

Radiographic Presentation	Differential Diagnosis Endodontic Lesion	Differential Diagnosis Non-Endodontic Lesion	Key Differentiating Features	Findings Warranting Histopathologic Evaluation
Well-defined periapical radiolucency associated with a non-vital tooth	Apical periodontitis, periapical granuloma, radicular cyst	Odontogenic keratocyst, central giant-cell granuloma	Non-endodontic lesions may occur despite positive vitality testing or minimal restorative history	Persistent lesion after adequate root canal treatment or atypical imaging features
Large (≥ 2 cm) radiolucency with cortical expansion	Radicular cyst	Ameloblastoma, odontogenic keratocyst, central giant-cell granuloma	Significant cortical expansion, multilocularity, root displacement, or aggressive growth pattern suggest non-endodontic pathology	Large expanding lesions or lesions with aggressive radiographic appearance

Table 2. Cont.

Radiographic Presentation	Differential Diagnosis Endodontic Lesion	Differential Diagnosis Non-Endodontic Lesion	Key Differentiating Features	Findings Warranting Histopathologic Evaluation
Ill-defined radiolucency with pain or swelling	Acute apical abscess	Osteomyelitis, malignancy, metastatic disease	Diffuse borders, paresthesia, rapid progression, systemic symptoms	Progressive or poorly defined lesions
Periapical radiolucency associated with a vital tooth	Uncommon for endodontic disease	Cemento-osseous dysplasia, odontogenic cysts, malignancy	Positive vitality testing strongly suggests non-endodontic origin	Lesions associated with vital teeth
Mixed radiolucent and radiopaque lesion	Healing lesion or condensing osteitis	Fibrous dysplasia, cemento-osseous dysplasia	Ground-glass appearance, multifocality, symmetrical expansion	Progressive mixed lesions
Predominantly radiopaque periapical lesion	Condensing osteitis	Idiopathic osteosclerosis, osteoma	Condensing osteitis is usually associated with pulpal inflammation or necrosis	Imaging findings inconsistent with pulpal status

It is important to emphasize that all specimens obtained during surgical removal of periapical lesions should be routinely submitted for histopathological examination.

3. Pathophysiological Mechanisms

3.1. Microbial Etiology

Bacteria are the primary microbial agents responsible for periapical lesions, as demonstrated by multiple systematic reviews [34,35], colonizing the necrotic pulp tissue and root canal systems and forming complex multispecies biofilms that promote the onset and progression of periapical lesions [36]. The most abundant bacterial groups in periapical lesions are Firmicutes, Bacteroidetes, Actinobacteria, and Proteobacteria, with dominant types including *Fusobacterium*, *Prevotella*, *Porphyromonas*, *Streptococcus*, *Parvimonas*, *Actinomyces*, and *Pseudoramibacter* spp. [36,37]. Obligate anaerobes predominate in primary infections, whereas both anaerobic and facultative species are common in post-treatment lesions [37].

These microorganisms interact synergistically within biofilms to enhance their virulence and resistance to host defenses and endodontic treatment [37,38]. Bacterial metabolites, including endotoxins such as lipopolysaccharides and lipoteichoic acid, that are commonly found in periapical lesions are directly associated with local inflammation, bone resorption, and accompanying clinical symptoms [39]. Gram-negative bacteria are more abundant in symptomatic lesions, and their metabolic pathways (e.g., amino acid degradation, flagellum assembly) are linked to tenderness and pain [40]. Although fungi, archaea, and viruses may be present in the root canal system, their pathogenic role is considered minor compared with that of bacteria [38].

3.2. Host Immune and Bone Resorption

Periapical diseases develop via a complex interaction between the host's immune and inflammatory responses and microbial infection. Initially, neutrophils and macrophages

are recruited to the periapical tissues, with neutrophils serving as the first line of defense and macrophages modulating lesion progression and bone remodeling [41,42]. T cells are central to the immunopathology of periapical lesions. On the one hand, Th1 cells release proinflammatory cytokines, such as interferon- γ and interleukin (IL)-12, which drive lesion progression and bone resorption. On the other hand, Th17 cells produce IL-17, which intensifies inflammation and is associated with symptomatic lesions [43]. In addition, Th2 and regulatory T cells produce anti-inflammatory cytokines (IL-10 and transforming growth factor- β), which contribute to lesion stability and limit tissue destruction [44].

The balance between these responses determines lesion activity. Bone resorption is primarily mediated by osteoclast activation through the RANKL/OPG signaling axis [45]. Proinflammatory cytokines (TNF- α , IL-1 β , IL-6), produced by immune cells in response to bacterial products, upregulate RANKL expression and promote osteoclast differentiation and activity [45,46]. A high RANKL/OPG ratio is the primary determinant of active bone destruction, whereas a low ratio is associated with quiescent or healing lesions.

Macrophages play a central role, both as sources of cytokines and as regulators of lesion progression, with their polarization status influencing whether lesions are stable or exhibit active bone resorption [42]. Lipopolysaccharide from endodontic pathogens directly stimulates macrophages to release IL-1 α and TNF- α , which upregulate matrix metalloproteinases and further promote bone resorption [46].

3.3. Biological Basis of Healing

The biological basis of healing in periapical lesions is the resolution of inflammation and restoration of the periapical tissue structure through coordinated immune modulation, cellular repair, and bone regeneration. Healing is initiated when the source of infection and immunogenic stimulus is eliminated, typically via root canal treatment, which allows the chronic inflammatory tissue to transition into reparative tissue [47].

The reduction in bacterial load decreases expression of proinflammatory mediators (TNF- α , IL-1 β , RANKL) and increases anti-inflammatory cytokines (IL-10), promoting resolution of inflammation and downregulation of osteoclast activity [48]. Furthermore, mesenchymal stem/progenitor cells are recruited to the periapical region, where they differentiate into osteoblasts and fibroblasts, supporting new bone and connective tissue formation [47]. The Wnt/ β -catenin signaling pathway is a key regulator of this process, as its activation accelerates bone formation and healing. In addition, a low RANKL/OPG ratio, driven by anti-inflammatory T-cell subsets, favors lesion stability and bone repair [45,49].

The healing outcome depends on the host's reparative capacity, local immune environment, and the effectiveness of treatment in reducing or eliminating root canal microorganisms, as well as the presence of pro-healing molecular signals [36,38]. Particularly, systemic health factors such as a history of diabetes, smoking, or cardiovascular disease can impair the microvasculature, bone turnover, and fibroblast function, thereby delaying or preventing periapical healing [50].

4. Clinical Management

4.1. Endodontic Treatment

The standard treatment for periapical lesions is conservative non-surgical root canal therapy, which includes meticulous cleaning, shaping, and disinfection of the root canal system, followed by obturation with gutta-percha and a sealer, which aims to reduce or eliminate microorganisms from the root canal system [51,52]. This approach is effective for both small and large lesions, including cyst-like lesions. Systematic review report pooled periapical healing rates of approximately 83–87% for primary root canal treatment under loose criteria, and 78–87% for non-surgical retreatment [53]. Even large cyst-like periapical

lesions (>10 mm) can achieve success rates of 82–89% with non-surgical management alone [54], and a retrospective study of 199 teeth with large apical periodontitis lesions demonstrated success rates of 89% under loose criteria [55]. For extensive or cystic lesions, adjunctive decompression techniques, such as orthograde aspiration, accelerate healing and reduce lesion size, leading to early resolution [52].

The bacteriological status of the root canal at the time of obturation has been investigated as a predictor of periapical healing following endodontic treatment. A landmark study demonstrated that complete periapical healing occurred in 94% of cases with negative cultures at the time of root filling, compared with 68% when bacteria were still detectable, a statistically significant difference [56]. A randomized clinical trial conducted by Zahran et al. [57] using quantitative PCR confirmed that an enhanced infection control protocol that reduced bacterial contamination before obturation resulted in three-fold higher odds of favorable 1-year outcomes. However, the predictive value of current canal sampling techniques has limitations, including false-positive and false-negative results, and the clinical utility of routine microbiological monitoring before obturation remains debated.

4.2. Indications for Surgical Treatment

Surgical intervention is indicated when adequate chemomechanical debridement and disinfection have failed to eliminate the infection, specifically: persistent or recurrent periapical pathology after adequate non-surgical root canal treatment or retreatment; anatomical factors preventing non-surgical management (such as inaccessible canals, complex root anatomy, or the presence of an intraradicular post that cannot be safely removed), large cystic lesions unresponsive to conservative decompression or intracanal medicaments and cases where biopsy is required to rule out non-endodontic or neoplastic pathology [58].

4.3. Surgical Treatment Modalities

The choice of modality depends on lesion size, location, anatomical factors, and the need for histopathological diagnosis. Apicoectomy (root-end resection) involves surgical removal of the apical portion of the root to eliminate persistent extraradicular infection and contaminated apical anatomy, followed by retrograde preparation and filling using a biocompatible calcium-silicate material such as mineral trioxide aggregate (MTA). This is often combined with periapical curettage to remove granulation or cystic tissue [58,59]. Modern endodontic microsurgery incorporates high magnification, ultrasonic root-end preparation, and minimal bevel angles, achieving success rates of 69–94% [60]. However, periapical curettage may be performed without root-end resection, particularly when the crown-to-root ratio is unfavorable. Adjunctive application of leukocyte- and platelet-rich fibrin to the bone defect enhances healing rates and reduces postoperative symptoms [59].

Enucleation is a surgical treatment modality in which the entire pathological lesion, including its lining, is carefully separated from the surrounding bone and removed in one piece, aiming for complete excision while preserving adjacent anatomical structures [61]. Spontaneous bone regeneration following enucleation is generally favorable, with studies reporting approximately 90% cavity healing by 12 months and progressive bone density increases over 24 months [62,63].

In contrast, marsupialization is a surgical technique in which a cystic lesion is opened, and its lining is sutured to the oral mucosa, creating a permanent window that allows continuous drainage and decompression of the cystic cavity. This reduces the intracystic pressure, promotes gradual shrinkage of the lesion, and facilitates bone regeneration within the defect. This technique is applied for the surgical management of extremely large odontogenic cysts, including radicular cysts and odontogenic keratocysts [64]. Treatment duration generally ranges from 1 to 2 years and requires consistent patient compliance with

irrigation; marsupialization most commonly serves as a preliminary treatment to reduce lesion size before definitive enucleation [64].

4.4. Assessment of Treatment Success

Treatment success is defined as the absence of clinical symptoms, such as pain, swelling, and tenderness to percussion or palpation; resolution of sinus tracts; restoration of normal function [59,65]; and complete resolution or significant reduction of the periapical lesion on imaging studies [65,66]. A volumetric change in radiolucency $\geq 20\%$ on CBCT is considered clinically significant for monitoring healing trends [67]. It should be noted that CBCT-based assessment using strict criteria (complete radiographic resolution) yields substantially lower success rates (36–45%) compared with loose criteria (85–87%), reflecting the higher sensitivity of CBCT in detecting residual radiolucencies that may not be clinically significant [68].

5. Biomarkers

5.1. Investigational Inflammatory Biomarkers

Several inflammatory biomarkers have been investigated as potential tools for assessing disease activity and predicting healing outcomes in apical periodontitis; however, none have been validated for routine clinical diagnostic use, and their application remains confined to research settings [69]. Inflammatory biomarkers associated with periapical lesions include proinflammatory cytokines such as IL-1 β , IL-6, IL-8, TNF- α , fibroblast growth factor-23, high-sensitivity C-reactive protein, asymmetric dimethylarginine (ADMA), matrix metalloproteinases (MMP-2, MMP-9), and chemokines (CCL-2, CXCL-8) [70–73]. Systematic reviews and meta-analyses have reported that levels of CRP, IL-6, and ADMA are significantly elevated in the peripheral blood of patients with apical periodontitis compared with healthy controls [74]. Similarly, elevated concentrations of TNF- α and IL-1 β have been detected in gingival crevicular fluid from teeth with apical periodontitis [73].

5.2. Limitations and Current Status

Despite promising research findings, several important limitations preclude the routine clinical application of biomarkers in the management of periapical lesions. The existing evidence is derived predominantly from cross-sectional studies and small cohorts, with high heterogeneity in sampling methods (serum, gingival crevicular fluid, periradicular tissue fluid), analytical techniques (ELISA—enzyme-linked immunosorbent assay, multiplex immunoassay, proteomic panels), and diagnostic criteria [69–74]. No standardized sampling protocols, validated reference ranges, or established clinical decision thresholds currently exist for any endodontic biomarker. Furthermore, cost-effectiveness data are lacking, and the laboratory infrastructure required for biomarker analysis is not available in routine dental practice settings.

At present, biomarker analysis is not part of routine clinical practice for the diagnosis or monitoring of periapical lesions. Biomarkers represent a promising area of ongoing research that may eventually complement conventional diagnostic methods, but their translation into validated, cost-effective clinical tools has not yet been achieved.

6. Future Perspectives

6.1. Artificial Intelligence

Artificial intelligence (AI) and deep learning models have demonstrated pooled sensitivity and specificity of 94% and 96%, respectively, for detecting periapical radiolucencies on radiographs and CBCT [75]. Furthermore, vision-language AI models have demonstrated promising performance in periapical lesion detection and treatment selection tasks [76].

Despite excellent diagnostic performance in experimental settings, the clinical generalizability, interpretability, and ethical integration of AI systems remain insufficiently validated, necessitating prospective multicenter studies before widespread clinical implementation.

6.2. *Advances in Imaging*

Dental-dedicated magnetic resonance imaging (ddMRI) is emerging as a radiation-free alternative, with a recent feasibility study reporting 88% accuracy for pulp vitality and 94% for apical periodontitis detection [77]. Additionally, second-generation CBCT systems based on carbon nanotube technology and stationary intraoral tomosynthesis are under development and may further enhance imaging capabilities while potentially reducing radiation dose [78].

Although emerging imaging modalities show considerable promise for improving diagnostic accuracy while reducing radiation exposure, their clinical applicability, cost-effectiveness, and long-term comparative benefits still require further validation.

6.3. *Targeted Personalized Therapies*

Immunotherapeutic strategies, including cytokine-targeted therapy, antioxidant therapy, and adoptive cell therapy, are being explored as adjuncts to conventional endodontic treatment to modulate the periapical inflammatory microenvironment and promote healing [79]. Nanotechnology-based approaches, such as microenvironment-responsive nanozyme composites that integrate infection control, immune modulation, and tissue regeneration, have shown promising results in preclinical models [80]. Furthermore, genetic insights into interleukin polymorphisms may eventually guide risk stratification and individualized treatment planning, although population-based validation studies are required before clinical implementation [81].

Personalized biologic and nanotechnology-based therapies represent a promising paradigm shift in endodontics; however, most evidence remains preclinical, and robust human trials are essential before translation into routine practice.

6.4. *Molecular Diagnostics and Genomics*

Advances in omics technologies (proteomics, genomics, transcriptomics) are beginning to elucidate the molecular basis of susceptibility to apical periodontitis. The first genome-wide association study of apical periodontitis identified novel risk loci and sex-specific genetic contributions to disease susceptibility [82]. MicroRNA profiling has revealed over 100 differentially expressed miRNAs in periapical lesions, with miR-10a-5p identified as a potential modulator of inflammation and healing [83].

Current molecular and genomic discoveries provide valuable insight into disease pathogenesis and individual susceptibility, yet their clinical relevance and reproducibility must be confirmed through large-scale translational research.

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Abbreviations

The following abbreviations are used in this manuscript:

CBCT	Cone-Beam Computed Tomography
FOV	Field Of View
IL	Interleukin
MMP	Matrix Metalloproteinases
ADMA	Asymmetric Dimethylarginine
ELISA	Enzyme-Linked Immunosorbent Assay
AI	Artificial Intelligence

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